ES S-138: Introduction to Probability for Engineering and Data Science **Summer 2020**

Lecture 5: Jul. 7, 2020

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Note: The topic of this lecture is expectation, linearity, variance, and multiple random variables. Related reading can be found in sections 2.4, 2.5, and 2.7 in the textbook.

Review: Common Random Variables

Name	Model	PMF
Bernoulli	Toss a biased coin	$P_X(x) = \begin{cases} p & \text{if } x = 1 \\ 1-p & \text{if } x = 0 \\ 0 & \text{o.w.} \end{cases}$
Geometric	Toss a coin until you see "H"	$P_X(k) = (1-p)^{k-1}p$
Binomial	Toss a coin n times. $X = \#$ of "H"	$P_X(k) = \binom{N}{k} p^k (1-p)^{N-k}$
Poisson	Like binomial, but set $n \to \inf$, $p \to 0$, $p = \frac{\lambda}{n}$, where λ is the average number of events	$P_X(k) = \frac{\lambda^k e^{-k}}{k!}$

5.1 Summary Statistics

The PMF of a random variable provides the complete details regarding the RV. In many cases, it would be useful to have more summarized information. Some examples include the maximum likely value of a PMF (i.e., the point with highest probability) or a (weighted) average of the possible values of the RV, X.

5.1.1 Expectation (Average Value)

Expectation of a RV is defined as the weighted average of the possible values of the RV, X. By using the corresponding probability of a given outcome value, we can weight each outcome. Hence we define the expectation of a RV X, with a given PMF $P_X(x)$ as

$$\underbrace{\mathbb{E}[X]}_{\text{expectation}} = \underbrace{\sum_x x \cdot P_X(x)}_{\text{weighted average}}$$

5.1.2 Variance

Another important summary quantity about the RV is the measure of the spread of its PMF. This quantity is important, for example, when differentiating two RVs with the same mean but with a different range of uncertainty surrounding the mean. *Variance* provides a measure of dispersion of a RV around its mean. To quantify variance, we first remove (subtract) the mean from the random variable itself, and get the expectation of the square of the difference:

$$\mathbf{var}(X) = \mathbf{E}\Big[(X - \mathbf{E}[X])^2\Big]$$

Note: since $(X - \mathbf{E}[X])^2$ can only take nonnegative values, the variance is always nonnegative.

The square root of variance is another measure of dispersion called **standard deviation** which is often easier to interpret because it shares the same units as the original RV, X; whereas, variance is in terms of square units. Standard deviation is denoted

$$\sigma_X = \sqrt{\mathrm{var}(X)}$$

Calculating variance could be done by using the definition of expected value, but we'd need to calculate the PMF of $(X - \mathbf{E}[X])^2$. That is,

$$\mathrm{var}(X) = \mathbf{E}[Z] = \sum_z z \cdot P_Z(z), \text{ where } Z = (X - \mathbf{E}[X])^2$$

Alternatively, we could use **the law of the unconscious statistician**.

5.1.3 The Law of The Unconscious Statistician (LOTUS)

Let X be a RV with PMF $P_X(x)$, and define $Y = g(X)$. The expectation of Y is

$$\mathbf{E}[Y] = \sum_y y \cdot P_Y(y) = \sum_x g(x) \cdot P_X(x)$$

To verify this "law," we would need to employ a method of weighting and grouping by mass, using the following:

$$P_Y(y) \quad = \quad \underbrace{\sum_{\{x|g(x)=y\}} P_X(x)}_{\text{summing masses } \forall\ x \text{ s.t. } g(x)=y}$$

$$\implies \mathbf{E}\big[g(x)\big] = \mathbf{E}[Y] = \sum_y \sum_{\{x|g(x)=y\}} g(x)P_X(x) = \sum_x g(x) \cdot P_X(x)$$

Note: For the purposes of ES138-S, we only expect you to recognize LOTUS and apply it when appropriate—i.e., know that for $Y = g(x)$, $\mathbf{E}(Y) = \sum_x g(x)P_X(x)$. The validation method might require some additional reading of B&T Chapter 2.

Example: Suppose that X is a uniform random variable in the set $\{-2, -1, 0, 1, 2\}$. Consider the random variable $Y = \max(X, 0)$. The expectation of Y can be expressed as follows

$$\begin{aligned}\mathbf{E}(Y) = \sum_y y \cdot P_Y(y) &= 0 \cdot P_Y(0) + 1 \cdot P_Y(1) + 2 \cdot P_Y(2) \\ &= 0 \cdot 3/5 + 1 \cdot 1/5 + 2 \cdot 1/5\end{aligned}$$

Alternatively, one can compute the expectation as follows

$$\begin{aligned}\mathbf{E}(Y) = \sum_x g(x) \cdot P_X(x) &= g(-2) \cdot P_X(-2) + g(-1) \cdot P_X(-1) + g(0) \cdot P_X(0) + g(1) \cdot P_X(1) + g(2) \cdot P_X(2) \\ &= 0 \cdot 1/5 + 0 \cdot 1/5 + 0 \cdot 1/5 + 1 \cdot 1/5 + 2 \cdot 1/5\end{aligned}$$

The two expressions give the same expected value.

5.2 Variance, revisited

With LOTUS now in our toolkit, we can consider a re-definition of variance. Previously, we defined variance:

$$\mathbf{var}(X) = \mathbf{E}\Big[(X - \mathbf{E}[X])^2\Big]$$

Using an intermediary RV, Z, where $Z = (X - \mathbf{E}[X])^2$, we suggested solving for $P_Z(z)$ to find $\text{var}(X)$:

$$\text{var}(X) = \mathbf{E}[Z] = \sum_z z \cdot P_Z(z)$$

Now, using LOTUS, we know

$$\mathbf{var}(X) = \mathbf{E}\Big[(X - \mathbf{E}[X])^2\Big] = \sum_x \big(x - \mathbf{E}[X]\big)^2 \cdot P_X(x)$$

Finally, we can perform a few more operations as follows, to get an alternative form (and quite useful form) for variance:

$$\begin{aligned}\mathbf{var}(X) &= \sum_x \big(x - \mathbf{E}[X]\big)^2 \cdot P_X(x) \\ &= \sum_x \Big(x^2 - 2x\mathbf{E}[X] + \big(\mathbf{E}[X]\big)^2\Big) P_X(x) \\ &= \underbrace{\sum_x x^2 P_X(x)}_{\mathbf{E}[X^2] \text{ by LOTUS}} - 2\mathbf{E}[X] \underbrace{\sum_x x P_X(x)}_{\text{def'n of } \mathbf{E}[X]} + \big(\mathbf{E}[X]\big)^2 \underbrace{\sum_x P_X(x)}_{=1} \\ &= \mathbf{E}[X^2] - \big(\mathbf{E}[X]\big)^2\end{aligned}$$

In summary,

$$\text{var}(X) = \mathbf{E}\Big[(X - \mathbf{E}[X])^2\Big] = \sum_x \big(x - \mathbf{E}[X]\big)^2 \cdot P_X(x) = \mathbf{E}[X^2] - \big(\mathbf{E}[X]\big)^2$$

Example: Finding the variance of a Bernoulli RV—we have previously seen that the mean of a Bernoulli RV is $\mathbf{E}[X] = p$. Now, if we use this information in the definition of the variance,

$$\text{var}(X) = \underbrace{\mathbf{E}[X^2]}_{=p,\text{ by LOTUS}} - \underbrace{\big(\mathbf{E}[X]\big)^2}_{=p^2} = p - p^2 = p(1-p)$$

or, we can likewise see:

$$\begin{aligned}
\text{var}(X) &= \mathbf{E}\big[(X - \mathbf{E}[X])^2\big] \\
&= \mathbf{E}\big[(X-p)^2\big] \\
&= \mathbf{E}\big[g(X)\big] \\
&= \sum_x g(x) P_X(x) \\
&= (1-p)^2 p + (0-p)^2(1-p) \\
&= p(1-p)
\end{aligned}$$

In Fig. 5.1 below, we observe the variance of Bernoulli RV with respect to the parameter p. Clearly, the most unpredictable coin is the fair one (i.e., when $p = \frac{1}{2}$), with a maximum variance!

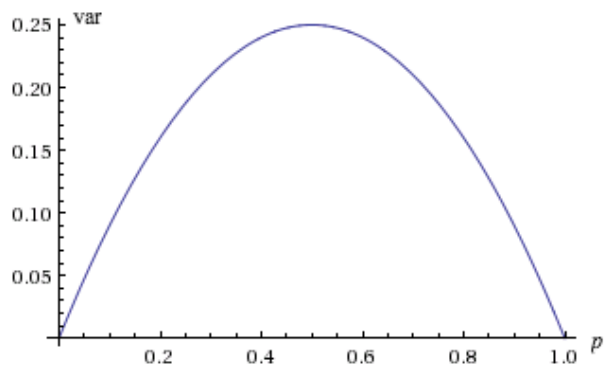


Figure 5.1: var(X) as a function of p

5.3 Properties of expectation and variance

$$\mathbf{E}[aX + b] = a\mathbf{E}[X] + b$$

Proof: We can prove the above statement by use of the definition of expectation.

$$\begin{aligned}
\mathbf{E}[aX+b] &= \sum_x (ax+b) P_X(x) \\
&= a\mathbf{E}[X] + b \cdot 1 \quad \blacksquare
\end{aligned}$$

$$\text{var}(aX + b) = a^2\text{var}(X)$$

Proof: Again, we can prove the above statement by use of the definition of expectation and a form of our definition of variance.

$$\begin{aligned}
\text{var}[aX + b] &= \mathbf{E}\big[(aX + b)^2\big] - \big[\mathbf{E}(aX + b)\big]^2 \\
&= \mathbf{E}[a^2X^2 + 2abX + b^2] - \big[a\mathbf{E}[X] + b\big]^2 \\
&= \mathbf{E}[a^2X^2 + 2abX + b^2] - a^2\mathbf{E}[X]^2 - 2ab\mathbf{E}[X] - b^2 \\
&= a^2\mathbf{E}(X^2) - a^2\mathbf{E}(X)^2, \text{ by linearity of expectation (see Section 5.5)} \\
&= a^2\text{var}(X) \ \blacksquare
\end{aligned}$$

5.4 Multiple Random Variables, Joint PMF

In general, when considering two discrete random variables X and Y, the **joint PMF** of the random variables X and Y is defined as follows

$$P_{X,Y}(x, y) = P(X = x \cap Y = y)$$

We can calculate the **marginal PMF** for X and Y by the following

$$P_X(x) = \sum_y P_{X,Y}(x, y) \qquad P_Y(y) = \sum_x P_{X,Y}(x, y)$$

We can show $P_X(x) = \sum_y P_{X,Y}(x, y)$ explicitly as

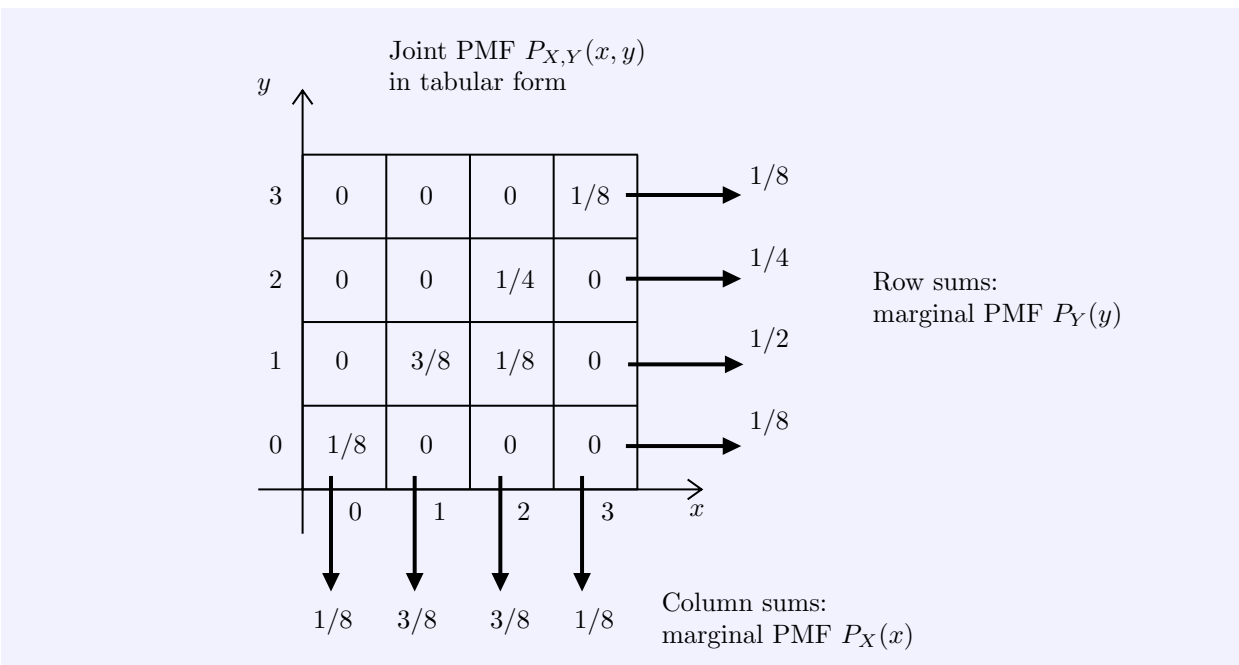
$$\begin{aligned}
P_X(x) &= \mathbb{P}\big(\{X = x\}\big) \\
&= \sum_y \mathbb{P}\big(\{X = x \ \cap \ Y = y\}\big) \\
&= \sum_y P_{X,Y}(x, y)
\end{aligned}$$

Example: Suppose you toss a coin three times. Define $X = \#$ of "Heads" you see in the span of the three tosses. Then define $Y =$ the length of the longest run of "Heads" you see (i.e., 0, 1, 2, or 3).

Let's first consider our sample space:

Ω	X	Y
HHH	3	3
HHT	2	2
HTH	2	1
HTT	1	1
THH	2	2
THT	1	1
TTH	1	1
TTT	0	0

From our sample space, we can easily construct our joint PMF in a tabular form, wherein (1) each entry contains the joint probability, (2) the sum of the rows gives the marginal PMF for Y, and (3) the sum of the columns gives the marginal PMF for X:



It follows from the above tabular method:

$$P_X(x) = \begin{cases} \frac{1}{8}, & x = 0 \\ \frac{3}{8}, & x = 1 \\ \frac{3}{8}, & x = 2 \\ \frac{1}{8}, & x = 3 \end{cases} \qquad P_Y(y) = \begin{cases} \frac{1}{8}, & y = 0 \\ \frac{4}{8}, & y = 1 \\ \frac{2}{8}, & y = 2 \\ \frac{1}{8}, & y = 3 \end{cases}$$

5.5 Linearity of Expectation

One important property of expectation is linearity:

$$\mathbf{E}[X_1 + X_2 + \cdots + X_N] = \mathbf{E}[X_1] + \mathbf{E}[X_2] + \cdots + \mathbf{E}[X_N]$$

Let us first show that for $Z = X + Y$, we have $\mathbf{E}[Z] = \mathbf{E}[X] + \mathbf{E}[Y]$. The idea is to use marginalization:

$$\begin{aligned}
\mathbf{E}[Z] &= \sum_x \sum_y (x+y) P_{XY}(x,y) \\
&= \sum_x \sum_y x \cdot P_{XY}(x,y) + \sum_x \sum_y y \cdot P_{XY}(x,y) \\
&= \sum_x x \Big(\sum_y P_{XY}(x,y) \Big) + \sum_y y \Big(\sum_x P_{XY}(x,y) \Big) \\
&= \sum_x x P_X(x) + \sum_y y P_Y(y), \text{ using marginalization} \\
&= \mathbf{E}[X] + \mathbf{E}[Y]
\end{aligned}$$

This property allows us to calculate the expectation of a random variable by decomposing it into a summation of several parts and calculate these parts individually. Usually, the expectation of these decomposed parts is easier to compute.

Example: Let's now find the mean of binomial distribution. We can easily redefine a binomial random variable as sum of N Bernoulli random variables such that each Bernoulli RV is denoted as

$$X_i = \begin{cases} 1, & \text{if True (or "Heads")} \\ 0, & \text{o.w.} \end{cases}$$

Our binomial RV composed of Bernoulli RVs, X_i:

$$X = X_1 + X_2 + \cdots X_N,$$

which is exactly equal to our "story" for the binomial (i.e., the number of "Heads" in a sequence of N tosses). Previously, we have seen that the Bernoulli RV has a mean (expected value) of p, so we conclude that for a bionomial RV, using the recently introduced property of linearity of expectation

$$\mathbf{E}[X] = \mathbf{E}[X_1] + \mathbf{E}[X_2] + \cdots \mathbf{E}[X_N] = Np$$

Example: Suppose that N people throw their hats into a box and then each picks up one hat at random. What is the expected value of X, where we define $X = \#$ of people who get back their own hat?

For the ith person, we can introduce a random (indicator) variable X_i that takes the value 1 if the person selects their own hat or takes the value 0 otherwise. For the first person to select a hat from the N hats, their probability of successfully picking their own hat is $P(X_1 = 1) = \frac{1}{N}$. What about any other? Considered alone, we could take the ith person as follows

$$P(X_i = 1) = \frac{(n-1)!}{n!} = \frac{1}{n}$$

That's $(n-1)!$ permutations for others' hats placements given person i has received their own hat over a total sample space of $n!$ distinct orderings.

Now, knowing that $P(X_i = 1) = \frac{1}{N}$ it is trivially known that $P(X_i = 0) = 1 - \frac{1}{N}$, so the expected value (mean) of X_i is

$$\mathbf{E}[X_i] = 1 \cdot \frac{1}{N} + 0 \cdot \left(1 - \frac{1}{N}\right) = \frac{1}{N}$$

X is defined as the number of people who get back their own hat, and X_i is an indicator variable for ith person, so we have

$$X = X_1 + X_2 + \cdots X_N,$$

$$\mathbf{E}[X] = \mathbf{E}[X_1] + \mathbf{E}[X_2] + \cdots \mathbf{E}[X_N] = N \cdot \frac{1}{N} = 1$$

5.6 Independent Random Variables

Recall: If two events, A and B, are independent, we have the following

$$P(A \cap B) = P(A) \cdot P(B)$$

$$P(A \mid B) = P(A) \text{ and } P(B \mid A) = P(B)$$

For random variables X and Y, we say they are independent if and only if for all pairs of $\{X = x\}, \{Y = y\}$

$$P_{X,Y}(x, y) = P_X(x) \cdot P_Y(y)$$

Equivalently, we can say

$$P_{X|Y}(x \mid y) = P_X(x) \text{ and } P_Y(y \mid x) = P_Y(y)$$

5.6.1 Properties of Independent Random Variables

If random variables X and Y are independent, then we have the following:

1. Functions of random variables $f(X)$ and $g(Y)$ are also independent
2. $\mathbf{E}[X \cdot Y] = \mathbf{E}[X] \cdot \mathbf{E}[Y]$;
3. $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$

$\implies$ for independent random variables $X_1, X_2, ..., X_n$, we can take the following:

$$\text{var}(X_1 + X_2 + \cdots + X_n) = \text{var}(X_1) + \text{var}(X_2) + \cdots + \text{var}(X_n)$$

Example: For random variable X which follows binomial distribution with parameters (n, p), find $\text{var}(X)$.

If we were to use the definition of variance to calculate $\text{var}(X)$ using the PMF of X, the process would be tedious (but doable). Conveniently, we can write the binomial RV as the summation of N independent Bernoulli random variables with parameter p, $X = X_1 + \cdots + X_N$. We know the variance of a Bernoulli random variable is $\text{var}(X_1) = \mathbf{E}[X_1^2] - \big(\mathbf{E}[X_1]\big)^2 = p(1-p)$. So $\text{var}(X)$ for our binomial RV is

$$\text{var}(X) = \text{var}(X_1) + \cdots + \text{var}(X_N) = N \cdot p(1-p)$$

Case of *dependent* Random Variables:
An easy example to show that the property $\text{var}(X+Y) = \text{var}(X) + \text{var}(Y)$ does not hold for dependent variables can be seen if you take a random variable X and a second RV, $Y = -X$. Clearly the two are not independent; likewise, $\text{var}(X+Y) = \text{var}(0) = 0 \neq 2\text{var}(X)$ ■

Last Updated by Dhilan Ramaprasad on August 17, 2020